

LAB3–COMPONENTMODELLING&ARCHITECTURAL PATTERN SELECTION

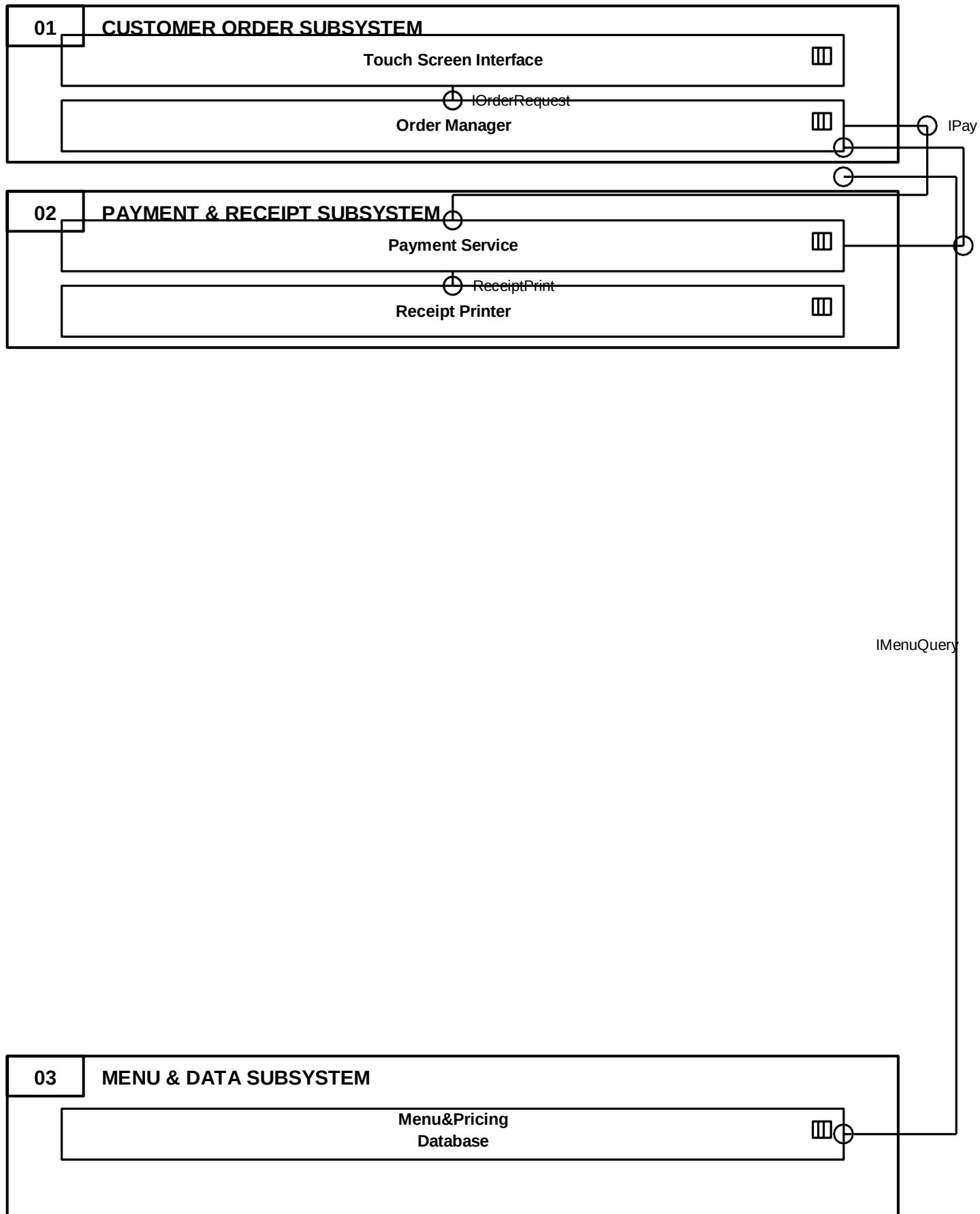
Final diagram in the same visual style as the reference you provided.

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Self-Service Coffee Kiosk – Component Diagram



Architecture Selection: Layered Architecture

We chose the Layered Architecture for the Self-Service Coffee Kiosk System. The first reason is clear separation of concerns. The system handles the touch-screen interface, customer orders, credit-card payments, menu and pricing information, and receipt printing. Separating these responsibilities makes the system easy to understand, maintain, test, and extend.

The second reason is the structured request-processing flow. A customer's request enters through the TouchScreen Interface and goes to the Order Manager. The Order Manager gets menu and pricing information from the Menu & Pricing Database, sends the payment request to the Payment Service, and sends the completed order details to the Receipt Printer.

Security Advantage: Payment processing is isolated in the Payment Service instead of being handled directly by the user interface. This reduces unnecessary exposure of payment-related information.

Performance Benefit: Menu and pricing information can be accessed efficiently through the data layer while the Order Manager handles the main business operations. This separation allows components to be optimized independently.

Therefore, the Layered Architecture provides a clear, modular, secure, and maintainable solution for the Self-Service Coffee Kiosk System.